

Dr. John Mwangi Wandeto

Senior Lecturer | AI Researcher | Software Development Consultant

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 ResearchGate

Profile

PhD in Computer Science (University of Strasbourg, France) with a career bridging classroom teaching (TSC professional recognition) and university academic leadership. Currently Senior Lecturer at Dedan Kimathi University of Technology (DeKUT), where I have served as Dean of the School of Computer Science and Information Technology and Director of the Software Development Center.

My pedagogical foundation—combined with doctoral research in AI—positions me at the intersection of *pedagogy and modern technology*. I train AI model to help students learn from course-specific materials. Creator of a commercially available AI Teaching Assistant, suitable for even 100+ student classes. Active researcher in medical image analysis, explainable AI, and generative models.

Products & Services

 **AI Teaching Assistant for Lecturers** A portable AI tutor that runs on the lecturer's laptop. Students access via browser. AI answers using YOUR course materials.

- **Zero internet required** — classroom Wi-Fi only
- **QR code attendance** with anti-cheating
- **Analytics dashboard** to track struggling students

Pricing: KSh 5,000 one-time setup (introductory)

 **Retail & Inventory Management System** Web-based POS and inventory system for small to medium retail businesses.

- **Works offline** — syncs when internet available
- **Real-time stock tracking** and low-stock alerts
- **Profit/loss analytics** by product or category
- **Supplier management** and purchase orders

 **Transport & Fleet Management** System for matatu/sacco operators, delivery services, and logistics companies.

- **Real-time vehicle tracking** (GPS integration)
- **Driver performance monitoring** (speeding, idling)
- **Fuel consumption analytics** to reduce costs
- **Route optimization** and passenger counting

 **Hospital Management System** Comprehensive system for clinics, nursing homes, and hospitals.

 **UrinalysisAI (Medical AI)** AI-driven medical image analysis for microscopic urine analysis.

- **Explainable AI** for medical diagnostics
- **NVIDIA-accelerated** for fast inference
- **Funding:** Kenya Education Network (KENET) 2023-2024

 **Smart Agriculture Platform** Farm management system for Kenyan agribusiness (tea, coffee, horticulture, dairy).

- **Crop cycle tracking** and input cost management
- **Weather integration** and harvest predictions
- **Mobile-first design** for farm workers
- **Group purchasing** (cooperative buying)

 **School Management System** Complete platform for primary/secondary schools (supports Kenyan curriculum).

- **Student registration** and fee tracking (M-Pesa)
- **Exam management** and KNEC-format gradebook
- **Parent/teacher communication** portal
- **Timetable and staff allocation**

 **SACCO & Microfinance Platform** Digital platform for Savings and Credit Cooperative Organizations (SACCOs).

- **Patient registration** and electronic medical records
- **Appointment scheduling** and queue management
- **Pharmacy and lab integration**
- **NHIF claim processing** support

 **Property Management System** Platform for landlords, real estate agents, and housing cooperatives.

- **Unit/tenant management** and lease tracking
- **Rent collection** and arrears reporting (M-Pesa)
- **Maintenance request** portal
- **Document management** (leases, inspections)

- **Member registration** and share tracking
- **Loan application** and approval workflow
- **Dividend calculation** and distribution
- **Mobile money integration** (M-Pesa)

 **Custom Business Applications** Tailored software solutions for specific business needs.

- **Data analytics dashboards** (sales, operations, HR)
- **Workflow automation** and approval systems
- **Document management** and digital signatures
- Fixed price or time-based engagement

Technical Expertise

Machine Learning & AI Problem-Solving I use modern machine learning concepts to define complex problems and engineer effective solutions across computer vision, generative AI, and edge deployment.

NVIDIA AI Stack

- NeMo (conversational AI)
- TensorRT-LLM (inference)
- CUDA acceleration
- Edge deployment

ML Frameworks

- PyTorch / TensorFlow
- LangChain / Ollama (RAG)
- YOLO (object detection)
- Self-Organizing Maps

Solution Domains

- Medical image analysis
- Explainable AI (XAI)
- Generative models
- AI for education

Full-Stack & DevOps

- FastAPI / Python · Docker / Kubernetes · PostgreSQL · GitHub Actions

Current Research & Funding

UrinalysisAI Project (KENET Grant 2023-2024)

- AI-driven medical image analysis for microscopic urine analysis
- YOLO-based object detection + LLaMA/Generative AI
- Focus: Small data optimization, edge deployment, explainable AI

Selected Publications

2025

- Naivasha, P., Musumba, G., Gikunda, P., *Wandeto, J.* “An Intelligent Market Model that Incorporates Individual Trader’s Behavior to Simulate Forex Trading Data.” **Computing Open**.

2023

- *Wandeto, J.M.*, Dresch-Langley, B. “SOM-QE ANALYSIS: A biologically inspired technique to detect and track meaningful changes within image regions.” **Software Impacts** (Elsevier).
- Liu, R., *Wandeto, J.*, et al. “Spatiotemporal Modeling of Grip Forces Captures Proficiency in Manual Robot Control.” **Bioengineering** (MDPI).
- Dresch-Langley, B., *Wandeto, J.M.* “Human Symmetry Uncertainty Detected by a Self-Organizing Neural Network Map.” **Symmetry**.

2021

- Moso, J.C., Cormier, S., Fouchal, H., Runz, C., *Wandeto, J.M.* “Anomaly Detection on Data Streams for Smart Agriculture.” **Agriculture**.

2019

- *Wandeto, J.M.*, Dresch-Langley, B. “The quantization error in a Self-Organizing Map as a contrast and colour specific indicator.” **Neural Networks** (Elsevier).

2017

- *Wandeto, J.M., Nyongesa, H., Rémond, Y., Dresp-Langley, B.* “Detection of small changes in medical and random-dot images.” **Informatics in Medicine Unlocked.**

Experience

Dedan Kimathi University of Technology (DeKUT) — Senior Lecturer 2023–Present (Transfer of service from TSC — career transition from classroom teaching to university academic leadership)

- Teaching: Backend Development, Simulation & Modeling, Front-End Web Development, Research Methods, E-Commerce, Object-Oriented Programming
- Research: AI in medical imaging, explainable AI, generative models
- Postgraduate supervision (PhD/MSc)

DeKUT — Dean, School of Computer Science and Information Technology 2019–2023

- Led academic programs, curriculum development, and faculty affairs
- Oversaw accreditation and quality assurance

DeKUT — Director, Software Development Center 2019–2023

- Founded SDC; built 15+ in-house applications
- Led MS Navision ERP customization team
- Coordinated industry partnerships and student internships

DeKUT — Assistant Lecturer / Chair of Department 2010–2019

- Led department to ISO certification
- Taught undergraduate and postgraduate courses

Teachers Service Commission (TSC No. 373037) — Trained Teacher Prior to 2010

- Active professional teaching qualification retained
- Recognition as a registered teacher (transfer of service to DeKUT)

Postgraduate Supervision (Computer Science)

PhD Completed (1)

- Data Stream Mining Approaches in Intelligent Transportation Systems and Precision Agriculture (2022)

PhD Defending (1)

- Modelling an Intelligent high-risk Foreign Exchange Market using Game Theory (2026)

MSc Completed (1)

- MULTIMODAL CYBERBULLYING DETECTION USING BERT AND RESNET50V2 (2025)

MSc Ongoing (2)

- (In progress)

Education

PhD in Computer Science — University of Strasbourg, France **2018** | Thesis: Self-Organizing Map Quantization Error Approach for Detecting Temporal Variations in Image Sets

Masters in Computer Applications — Kenyatta University, Kenya **2009** | Research: Using ICT to Enhance Learning of Physics in Kenyan Secondary Schools

B.Ed. (Physics/Mathematics) — Egerton University, Kenya **1996** | Registered Teacher (TSC No. 373037 — professional recognition)

Academic Service

- **Reviewer:** Frontiers in Neurorobotics, BMC Psychology, Elsevier (Computers in Biology and Medicine), Frontiers in Psychology
- **Curriculum Developer:** MSc/PhD Computer Science programs at DeKUT
- **Peer Review:** 2 reviews for 2 publications/grants (verified by ORCID)

Languages

🇬🇧 English — **Fluent**

🇫🇷 French — **Working knowledge**

🇰🇪 Swahili — **Native**

Professional Affiliations

- Teachers Service Commission (TSC No. 373037) — Verified by TSC (Professional Recognition)
- ORCID: 0000-0002-7502-2517

Referees

Prof. Ndirangu Kioni Professor of Mechanical Engineering Dedan Kimathi University of Technology ✉️ ndirangukioni[at]dkut.ac.ke 📞 +254 722 773095

Dr. Birgitta Dresp-Langley (HDR) CNRS Research Director / Directeur de Recherche CNR University of Strasbourg ✉️ birgitta.dresp[at]unistra.fr



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